

Scenario 1 Results: Hotspot for Network Traffic Analysis and Control

Figure 1 shows the real-time association of devices with their respective IP and MAC addresses, as well as the interface to which they are connected.

```
Every 1.0s: arp -a
Xiaomi-11-Lite-5G-NE (10.0.1.12) at 46:92:14:a3:91:b0 [ether] on wlan0
LAPTOP-17BJFRV9.mshome.net (192.168.137.1) at 98:fa:9b:5c:dd:4a [ether] on eth0
```

Figure 1. Connection from user to hotspot.

Figure 2 shows multiple connected users, IP addresses are visible when the device cannot be recognized; otherwise, the make, model, or name assigned by the user will be displayed.

```
spn@raspberrypi:~ $ con
Address          HWtype  HWaddress      Flags Mask    Iface
157.148.63.205   (incomplete)          wlan0
129.226.3.47     (incomplete)          wlan0
183.47.125.78    (incomplete)          wlan0
101.32.104.4     (incomplete)          wlan0
123.151.56.102   (incomplete)          wlan0
101.32.104.41    (incomplete)          wlan0
101.32.118.25    (incomplete)          wlan0
Xiaomi-11-Lite-5G-NE ether    5a:de:f7:63:2f:96 C    wlan0
120.241.190.60   (incomplete)          wlan0
121.51.162.69    (incomplete)          wlan0
^C
spn@raspberrypi:~ $ fcon
? (157.148.63.205) at <incomplete> on wlan0
? (129.226.3.47) at <incomplete> on wlan0
? (183.47.125.78) at <incomplete> on wlan0
? (101.32.104.4) at <incomplete> on wlan0
? (123.151.56.102) at <incomplete> on wlan0
? (101.32.104.41) at <incomplete> on wlan0
? (101.32.118.25) at <incomplete> on wlan0
Xiaomi-11-Lite-5G-NE (10.0.1.19) at 5a:de:f7:63:2f:96 [ether] on wlan0
? (120.241.190.60) at <incomplete> on wlan0
? (121.51.162.69) at <incomplete> on wlan0
dns.google (8.8.8.8) at <incomplete> on wlan0
dns.google (8.8.4.4) at <incomplete> on wlan0
? (43.154.240.84) at <incomplete> on wlan0
? (175.27.38.131) at <incomplete> on wlan0
? (109.244.172.79) at <incomplete> on wlan0
raspberrypi.mshome.net (192.168.137.232) at <incomplete> on wlan0
? (116.130.223.103) at <incomplete> on wlan0
LAPTOP-17BJFRV9.mshome.net (192.168.137.1) at <incomplete> on wlan0
? (43.130.30.240) at <incomplete> on wlan0
```

Figure 2. Connecting multiple users to hotspot.

Figure 3 presents real-time traffic capture, showing that all traffic passing through the hotspot is intercepted, allowing the identification of visited domains and device IP addresses.

```

epn@raspberrypi:~$ $ capa
Running as user "root" and group "root". This could be dangerous.
Capturing on 'wlan0'
1 0.000000000 10.0.1.12 -> 10.0.1.1 DNS 74 Standard query 0x39e6 A www.google.com
2 0.001839634 10.0.1.1 -> 10.0.1.12 DNS 104 Standard query response 0x39e6 A www.google.com A 142.250.78.4
3 6.064943115 10.0.1.12 -> 10.0.1.1 DNS 95 Standard query 0x6549 A suggestqueries-clients6.youtube.com
4 6.080251543 10.0.1.12 -> 10.0.1.1 DNS 73 Standard query 0x2f6c A m.youtube.com
5 6.081847792 10.0.1.1 -> 10.0.1.12 DNS 102 Standard query response 0x2f6c A m.youtube.com A 172.217.173.206
6 6.085376904 10.0.1.1 -> 10.0.1.12 DNS 146 Standard query response 0x6549 A suggestqueries-clients6.youtube.com A 142.251.135.
174
7 6.295498600 10.0.1.12 -> 10.0.1.1 DNS 75 Standard query 0xa442 A www.youtube.com
8 6.297478807 10.0.1.1 -> 10.0.1.12 DNS 406 Standard query response 0xa442 A www.youtube.com CNAME youtube-ui.l.google.com A 14
2.250.218.142 A 142.251.132.78 A 142.251.132.110 A 142.251.132.142 A 142.251.135.174 A 172.217.28.110 A 172.217.30.206 A 172.217.172.14 A
172.217.173.46 A 172.217.173.206 A 142.250.78.14 A 142.250.78.46 A 142.250.78.78 A 142.250.78.110 A 142.250.78.142 A 142.250.78.174
9 17.439049944 10.0.1.12 -> 10.0.1.1 DNS 73 Standard query 0xc119 A yt3.ggpht.com
10 17.458656021 10.0.1.1 -> 10.0.1.12 DNS 184 Standard query response 0xc119 A yt3.ggpht.com CNAME photos-ugc.l.googleusercontent.com A 142.250.78.33
11 28.821112021 10.0.1.12 -> 10.0.1.1 DNS 75 Standard query 0x580b A play.google.com
12 28.843241685 10.0.1.1 -> 10.0.1.12 DNS 106 Standard query response 0x580b A play.google.com A 142.250.78.78
13 30.451031807 10.0.1.12 -> 10.0.1.1 DNS 93 Standard query 0x211e A firebaselogging-pa.googleapis.com
14 30.471416888 10.0.1.1 -> 10.0.1.12 DNS 366 Standard query response 0x211e A firebaselogging-pa.googleapis.com A 142.250.78.10
A 142.250.78.42 A 142.250.78.74 A 142.250.78.106 A 142.250.78.138 A 142.250.78.170 A 142.250.218.106 A 142.250.218.138 A 142.251.132.74
A 142.251.132.106 A 142.251.132.138 A 142.251.135.170 A 172.217.28.106 A 172.217.30.202 A 172.217.172.10
15 63.881130552 10.0.1.12 -> 10.0.1.1 DNS 71 Standard query 0x9077 A i.ytimg.com
16 63.883803360 10.0.1.1 -> 10.0.1.12 DNS 306 Standard query response 0x9077 A i.ytimg.com A 142.250.218.118 A 142.250.218.150 A
142.251.132.86 A 142.251.132.118 A 142.251.132.150 A 142.251.135.182 A 172.217.30.214 A 172.217.172.22 A 142.250.78.22 A 142.250.78.54 A
142.250.78.86 A 142.250.78.118 A 142.250.78.150 A 142.250.78.182
17 65.979796094 10.0.1.12 -> 10.0.1.1 DNS 73 Standard query 0x3c64 A m.youtube.com
18 66.003390012 10.0.1.1 -> 10.0.1.12 DNS 102 Standard query response 0x3c64 A m.youtube.com A 142.251.135.174

```

Figure 3. Capture traffic from the entire network.

Results of Scenario 2: Phishing through a Cloned Web

Figure 4 shows the execution of the Ngrok service and the public address provided, as well as the users who have accessed the cloned web.

```

ngrok
K8s Gateway API support available now: https://ngrok.com/r/k8sgb

Session Status      online
Account             prometheus.dexm@gmail.com (Plan: Free)
Version             3.10.1
Region              South America (sa)
Latency             165ms
Web Interface       http://127.0.0.1:4040
Forwarding           https://e942-157-100-135-40.ngrok-free.app -> http://localhost:8000

Connections         ttl    opn    rt1    rt5    p50    p90
                   32     0      0.28   0.09   0.44   5.06

HTTP Requests
-----
POST /ajax/bz                404 Not Found
POST /ajax/webstorage/process_keys/ 404 Not Found
POST /ajax/bz                404 Not Found
POST /ajax/bz                404 Not Found
POST /ajax/bz                404 Not Found
GET /Facedemo_files/hsts-pixel.gif 200 OK
GET /rsrc.php/v3/yB/r/Y0L6f5sxdIV.png 404 Not Found
GET /rsrc.php/v3/yU/r/07neImd9XSI.png 404 Not Found
GET /Facedemo_files/sKtrEJAtiUM.js 200 OK
GET /Facedemo_files/ui2DkP-wt 7.js 200 OK

```

Figure 4. Public IP and connection to cloned page using ngrok.

Figure 5 shows the cloned page accessible from the public link generated by the Ngrok service.

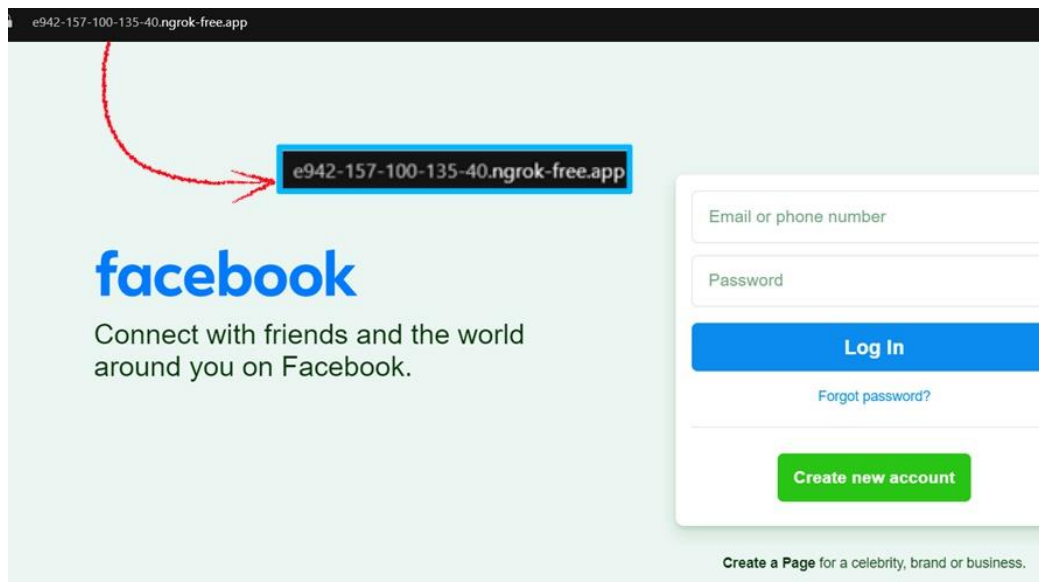


Figure 5. Access to cloned page via ngrok.

Figure 6 shows the capture of HTTP POST events generated by credential entry on the cloned page.

```
epn@raspberrypi:~ $ post
0020 a2 83 a9 bb 08 c1 0c 97 0f 04 a6 86 7a 3a 08 66 .....z:.f
0030 64 9c e2 00 00 00 6f 01 00 00 00 6a 00 0f 01 00 d.....o....j....
0040 00 00 63 01 00 09 3a 80 00 00 00 00 00 3d 00 0c ..c.....=.
0050 f6 59 df ec e6 b8 0b 70 c7 ce 32 44 00 48 5d 74 .Y.....p...2D.H]t
0060 2a e4 66 1c d9 52 f6 63 04 e6 69 7a e2 ce bc b9 *.f..R.c..iz....
0070 7a 30 27 a2 df 26 30 f4 10 1a 83 a6 38 2a a8 de z0'..&0.....8*..
0080 eb 2e f1 7a 90 4a 7f a4 d1 e6 d3 86 78 b6 4d fa ...z.J.....X.M.
0090 08 c3 7e 62 d5 ef 24 df 8c da b7 03 e0 6f 4a 3f ...~b..$......oJ?
00a0 7c cd 77 ab c9 9b 19 f1 04 00 24 97 72 1c c2 de |.w.....$.r...
00b0 09 fd 15 be 73 d1 61 dd d2 12 ca 43 e4 15 da 16 ...s.a....C....
00c0 b6 66 af 63 69 10 fc dd e5 07 f4 d1 65 05 8c 17 .f.ci.....e...
00d0 f1 04 01 6e ab 1a 4b 8a 0c f6 94 ef 0b 40 1b 18 ...n..K.....@..
00e0 e3 f4 62 fe 81 48 2c 7c d0 d0 66 56 c8 82 d2 77 ..b..H,|..fv...w
00f0 00 d7 80 20 2c 29 d2 14 4b 02 9b 96 95 15 b6 17 ... ,)..K.....
0100 69 41 96 7d 0a 77 01 10 ac 06 ae a5 26 04 58 cb iA.}.w.....&.X.
0110 d0 97 9b 0f 51 38 5e 20 6d bc 5f 35 6b c0 c4 6c ....Q8^ m._5k..l
0120 4d 11 04 61 81 2d a0 96 50 6e ca 74 12 ed 46 2d M..a...Pn.t..F-
0130 ca 0d ae 49 a3 a3 6e f5 26 97 7d 39 01 1c 9a d1 ...I..n.&.)9....
0140 d9 e4 f9 a9 91 45 f9 fc 50 cc eb f2 5b 42 57 5d .....E..P...[BW]
0150 f2 ce bf db dd b4 85 96 59 1f 6c f1 71 38 9f e7 .....Y..l.q8..
0160 24 b2 8b 6e ba 46 12 dc 42 0c 05 3a 2e 93 5b d0 $.n.F..B.....[.
0170 0d 0f cf e3 c4 82 6c 28 eb 6e c4 61 e0 4d d0 97 .....l(.n.a.M..
0180 22 b9 a3 4e eb 96 11 15 bd fd e7 16 43 e3 be e2 "...N.....C...
0190 03 a6 86 ba 92 8e 3c 8f 68 0f 8b f9 51 86 6c 4d .....<.h...Q.lM
01a0 a7 8d 3d cd e4 0c e2 a1 5c 45 28 8b ae f7 ff 52 ..=.....\E(...R
01b0 0f 67 97 38 20 f5 6b e9 b0 d4 f3 a1 1a 11 d9 4e .g.8 .k.....N
01c0 11 61 01 a5 f0 ed e5 0a ad db e0 bf 22 ea ff 61 .a.....".a
01d0 09 9c f6 4a 0b 20 36 61 da 4b 04 94 8a 75 64 20 ...J. 6a.K...ud
01e0 52 74 7d ea 71 1f 13 8e 97 13 15 69 59 c7 23 8a Rt}.q.....iY.#.
01f0 5f ac 2d 24 57 57 d8 6c 7f 0d 2c 7b 43 c9 83 55 .-$wW..l...{C..U
```

Figure 6. Capture POST events.

Scenario 3 Results: Brute Force Attack for Password Cracking

Figure 7 shows the iptables firewall rules, initially allowing all traffic.

```
epn@raspberrypi:~ $ vt
[sudo] password for epn:
Chain INPUT (policy ACCEPT)
target    prot opt source                destination

Chain FORWARD (policy ACCEPT)
target    prot opt source                destination
ACCEPT    all  --  anywhere              anywhere             state RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
```

Figure 7. Display of iptables rules.

Figure 8 shows the use of the `netstop` command to stop the network, adding the `DROP all anywhere` rules, blocking all traffic.

```
epn@raspberrypi:~ $ netstop
epn@raspberrypi:~ $ vt
Chain INPUT (policy ACCEPT)
target    prot opt source                destination
DROP      all  --  anywhere              anywhere

Chain FORWARD (policy ACCEPT)
target    prot opt source                destination
DROP      all  --  anywhere              anywhere
ACCEPT    all  --  anywhere              anywhere             state RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
DROP      all  --  anywhere              anywhere
```

Figure 8. Network stop and display of iptables rules.

Figure 9 shows the compression of a file with a password, accompanied by a menu that makes it easy to select the range and type of characters used to configure the password complexity.

```
epn@raspberrypi:~ $ zp
Ingrese el rango de caracteres (2-4): 4
Ingrese el tipo de caracteres:
    1. Solo minúsculas
    2. Solo mayúsculas
    3. Solo números
    4. Todos los tipos: 1
Ingrese la contraseña:
adding: foto.jpg (deflated 23%)
```

Figure 9. Compression for brute force attack.

Figure 10 shows the onset of the brute-force attack to crack the password, and the proof of all possible character combinations is displayed.

```
epn@raspberrypi:~ $ dzp
found file 'foto.jpg', (size cp/uc 695469/898798, flags 9, chk 9036)
possible pw found: aaep ()
possible pw found: aafU ()
possible pw found: aai$ ()
possible pw found: aalJ ()
possible pw found: aaoX ()
possible pw found: aaqX ()
```

Figure 10. Brute force decompression

Figure 11 confirms the success of the attack, showing the password found and the time required.

```
possible pw found: ~~FC ()
possible pw found: ~~LQ ()
possible pw found: ~~02 ()
possible pw found: ~~0= ()
possible pw found: ~~St ()
possible pw found: ~~1) ()
possible pw found: ~~2B ()
possible pw found: ~~&x ()
possible pw found: ~~/G ()
possible pw found: ~~(X ()
possible pw found: ~~]* ()
possible pw found: ~~}N ()
possible pw found: ~~+H ()

*****
* Contraseña: casa *
*****

Complete: 0 minuto(s) 41 segundos.
```

Figure 11. Password found in brute force attack

Table 1 details the decryption times for different four-character passwords.

Table 1. Brute force decryption time.

Word (4 characters – all types)	Time
casa	1 min 29 seg
dedo	1 min 56 seg
esta	2 min 29 seg
fido	2 min 45 seg
gaza	3 min 6 seg

<i>hito</i>	3 min 33 seg
<i>isla</i>	4 min 10 seg
<i>jota</i>	4 min 27 seg
<i>kilo</i>	4 min 52 seg
<i>kiwi</i>	4 min 52 seg
<i>lado</i>	5 min 12 seg
<i>lata</i>	5 min 13 seg
<i>lazo</i>	5 min 16 seg

Figure 15, Figure 16 and Figure 17 show the iperf tests for the eth0 and wlan0 interfaces. The tests were carried out for both TCP and UDP, with an interval of 5 seconds and a total duration of 15 seconds.

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether b8:27:eb:c4:8f:3d brd ff:ff:ff:ff:ff:ff
    inet 192.168.137.12/24 brd 192.168.137.255 scope global dynamic noprefixroute eth0
        valid_lft 602982sec preferred_lft 451782sec
    inet6 fe80::3b7:3f3d:1df5:bf3/64 scope link
        valid_lft forever preferred_lft forever
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether b8:27:eb:91:da:68 brd ff:ff:ff:ff:ff:ff
    inet 10.0.1.1/24 brd 10.0.1.255 scope global wlan0
        valid_lft forever preferred_lft forever
    inet 169.254.3.176/16 brd 169.254.255.255 scope global noprefixroute wlan0
        valid_lft forever preferred_lft forever
    inet6 fe80::baa8:e7e5:3d94:65bd/64 scope link
        valid_lft forever preferred_lft forever
```

Figure 15. Interface address identification.

```
eprn@raspberrypi:~ $ iperf -c 192.168.137.12 -i 5 -t 15
-----
Client connecting to 192.168.137.12, TCP port 5001
TCP window size: 2.50 MByte (default)
-----
[ 3] local 192.168.137.12 port 58134 connected with 192.168.137.12 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 3] 0.0000-5.0000 sec  2.19 GBytes  3.77 Gbits/sec
[ 3] 5.0000-10.0000 sec  2.22 GBytes  3.81 Gbits/sec
[ 3] 10.0000-15.0000 sec  2.21 GBytes  3.79 Gbits/sec
[ 3] 15.0000-15.0009 sec   640 KBytes  5.76 Gbits/sec
[ 3] 0.0000-15.0009 sec  6.62 GBytes  3.79 Gbits/sec
eprn@raspberrypi:~ $ iperf -c 192.168.137.12 -u -i 5 -t 15
-----
Client connecting to 192.168.137.12, UDP port 5001
UDP buffer size: 176 KByte (default)
-----
[ 3] local 192.168.137.12 port 49095 connected with 192.168.137.12 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 3] 0.0000-5.0000 sec   639 KBytes   1.05 Mbits/sec
[ 3] 5.0000-10.0000 sec   640 KBytes   1.05 Mbits/sec
[ 3] 10.0000-15.0000 sec   640 KBytes   1.05 Mbits/sec
[ 3] 0.0000-15.0061 sec  1.88 MBytes   1.05 Mbits/sec
```

Figure 16. Try iperf eth0 interface.

```
epn@raspberrypi:~ $ iperf -c 169.254.3.176 -i 5 -t 15
-----
Client connecting to 169.254.3.176, TCP port 5001
TCP window size: 2.50 MByte (default)
-----
[ 3] local 169.254.3.176 port 38374 connected with 169.254.3.176 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0000-5.0000 sec  2.21 GBytes  3.79 Gbits/sec
[ 3] 5.0000-10.0000 sec  2.24 GBytes  3.85 Gbits/sec
[ 3] 10.0000-15.0000 sec  2.23 GBytes  3.83 Gbits/sec
[ 3] 15.0000-15.0010 sec   512 KBytes  4.32 Gbits/sec
[ 3] 0.0000-15.0010 sec  6.68 GBytes  3.82 Gbits/sec
epn@raspberrypi:~ $ iperf -c 169.254.3.176 -u -i 5 -t 15
-----
Client connecting to 169.254.3.176, UDP port 5001
UDP buffer size: 176 KByte (default)
-----
[ 3] local 169.254.3.176 port 51224 connected with 169.254.3.176 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0000-5.0000 sec   639 KBytes  1.05 Mbits/sec
[ 3] 5.0000-10.0000 sec   640 KBytes  1.05 Mbits/sec
[ 3] 10.0000-15.0000 sec   640 KBytes  1.05 Mbits/sec
[ 3] 0.0000-15.0061 sec  1.88 MBytes  1.05 Mbits/sec
```

Figure 17. Try iperf wlan0 interface.

Figure 18 shows the test of connecting devices to the hotspot by simulating 10 connections, setting a time of 5 seconds per connection, and displaying the information in gigabytes.

```
epn@raspberrypi:~ $ iperf -c 10.0.1.1 -P 10 -f g -t 5
-----
Client connecting to 10.0.1.1, TCP port 5001
TCP window size: 0.002 GByte (default)
-----
[ 3] local 10.0.1.1 port 56424 connected with 10.0.1.1 port 5001
[ 5] local 10.0.1.1 port 56446 connected with 10.0.1.1 port 5001
[ 8] local 10.0.1.1 port 56456 connected with 10.0.1.1 port 5001
[ 4] local 10.0.1.1 port 56434 connected with 10.0.1.1 port 5001
[ 9] local 10.0.1.1 port 56470 connected with 10.0.1.1 port 5001
[ 7] local 10.0.1.1 port 56476 connected with 10.0.1.1 port 5001
[10] local 10.0.1.1 port 56474 connected with 10.0.1.1 port 5001
[ 6] local 10.0.1.1 port 56472 connected with 10.0.1.1 port 5001
[11] local 10.0.1.1 port 56490 connected with 10.0.1.1 port 5001
[12] local 10.0.1.1 port 56500 connected with 10.0.1.1 port 5001
[ ID] Interval      Transfer    Bandwidth
[12] 0.0000-5.0152 sec  0.183 GBytes  0.314 Gbits/sec
[ 6] 0.0000-5.0118 sec  0.201 GBytes  0.344 Gbits/sec
[ 8] 0.0000-5.0293 sec  0.196 GBytes  0.334 Gbits/sec
[11] 0.0000-5.0014 sec  0.192 GBytes  0.329 Gbits/sec
[ 5] 0.0000-5.0003 sec  0.214 GBytes  0.367 Gbits/sec
[10] 0.0000-5.0160 sec  0.206 GBytes  0.353 Gbits/sec
[ 7] 0.0000-5.0298 sec  0.219 GBytes  0.374 Gbits/sec
[ 4] 0.0000-5.0004 sec  0.206 GBytes  0.354 Gbits/sec
[ 9] 0.0000-5.0105 sec  0.216 GBytes  0.370 Gbits/sec
[ 3] 0.0000-5.0265 sec  0.223 GBytes  0.380 Gbits/sec
[SUM] 0.0000-5.0269 sec  2.05 GBytes  3.51 Gbits/sec
[ CT] final connect times (min/avg/max/stddev) = 0.236/0.511/1.401/0.360 ms (tot/err) = 10/0
```

Figure 18. Connection simulation.

Figure 19 shows the variants resulting from the repetition of the test for 15, 20, 30, 50, 100, 150 and 210 simulated connections with iperf using the interface corresponding to the hotspot.

```
[ 3] local 10.0.1.1 port 42370 connected with 10.0.1.1 port 5001
[ ID] Interval      Transfer      Bandwidth
[  8] 0.0000-5.0065 sec  0.141 GBytes  0.242 Gbits/sec
[ 16] 0.0000-5.0093 sec  0.137 GBytes  0.235 Gbits/sec
[ 11] 0.0000-5.0179 sec  0.123 GBytes  0.210 Gbits/sec
[  6] 0.0000-5.0065 sec  0.155 GBytes  0.266 Gbits/sec
[ 14] 0.0000-5.0115 sec  0.135 GBytes  0.232 Gbits/sec
[  4] 0.0000-5.0004 sec  0.133 GBytes  0.229 Gbits/sec
[ 15] 0.0000-5.0004 sec  0.154 GBytes  0.264 Gbits/sec
[  3] 0.0000-5.0423 sec  0.124 GBytes  0.211 Gbits/sec
[ 17] 0.0000-5.0002 sec  0.139 GBytes  0.239 Gbits/sec
[  5] 0.0000-5.0147 sec  0.127 GBytes  0.217 Gbits/sec
[  7] 0.0000-5.0110 sec  0.143 GBytes  0.245 Gbits/sec
[ 10] 0.0000-5.0015 sec  0.121 GBytes  0.207 Gbits/sec
[  9] 0.0000-5.0102 sec  0.128 GBytes  0.220 Gbits/sec
[ 12] 0.0000-5.0003 sec  0.138 GBytes  0.237 Gbits/sec
[ 13] 0.0000-5.0161 sec  0.145 GBytes  0.248 Gbits/sec
[SUM] 0.0000-5.0338 sec  2.04 GBytes  3.48 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.213/0.378/0.784/0.276 ms (tot/err) = 15/0

[SUM] 0.0000-5.0164 sec  2.04 GBytes  3.49 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.211/0.394/1.253/0.245 ms (tot/err) = 20/0

[SUM] 0.0000-5.0686 sec  2.03 GBytes  3.44 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.206/0.385/1.350/0.366 ms (tot/err) = 30/0

[SUM] 0.0000-5.0935 sec  1.86 GBytes  3.13 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.195/0.331/2.309/0.328 ms (tot/err) = 50/0

[SUM] 0.0000-5.1969 sec  1.59 GBytes  2.63 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.192/0.354/2.872/0.300 ms (tot/err) = 100/0

[SUM] 0.0000-5.0883 sec  1.58 GBytes  2.67 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.189/0.351/9.781/0.788 ms (tot/err) = 150/0

[SUM] 0.0000-5.2963 sec  1.49 GBytes  2.42 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.191/0.419/13.188/1.141 ms (tot/err) = 210/0
```

Figure 19. Repetition of the test.

Figure 20 shows the maximum number of iperf-simulated connections supported by the Promedex device.

```

[120] 0.0000-5.1016 sec 0.007 GBytes 0.011 Gbits/sec
[ 22] 0.0000-5.1028 sec 0.007 GBytes 0.012 Gbits/sec
[135] 0.0000-5.0905 sec 0.008 GBytes 0.013 Gbits/sec
[ 26] 0.0000-5.1042 sec 0.008 GBytes 0.014 Gbits/sec
[ 23] 0.0000-5.0884 sec 0.005 GBytes 0.009 Gbits/sec
[ 30] 0.0000-5.1238 sec 0.007 GBytes 0.012 Gbits/sec
[202] 0.0000-5.0132 sec 0.006 GBytes 0.011 Gbits/sec
[198] 0.0000-5.0236 sec 0.010 GBytes 0.018 Gbits/sec
[170] 0.0000-5.0730 sec 0.006 GBytes 0.011 Gbits/sec
[ 32] 0.0000-5.0409 sec 0.006 GBytes 0.011 Gbits/sec
[ 69] 0.0000-5.0608 sec 0.006 GBytes 0.011 Gbits/sec
[ 62] 0.0000-5.1334 sec 0.008 GBytes 0.013 Gbits/sec
[117] 0.0000-5.1608 sec 0.008 GBytes 0.013 Gbits/sec
[195] 0.0000-5.1049 sec 0.007 GBytes 0.011 Gbits/sec
[157] 0.0000-5.0998 sec 0.008 GBytes 0.014 Gbits/sec
[ 39] 0.0000-5.0755 sec 0.013 GBytes 0.022 Gbits/sec
[ 89] 0.0000-5.1431 sec 0.016 GBytes 0.027 Gbits/sec
[ 21] 0.0000-5.1225 sec 0.008 GBytes 0.014 Gbits/sec
[124] 0.0000-5.0031 sec 0.005 GBytes 0.008 Gbits/sec
[204] 0.0000-5.0378 sec 0.007 GBytes 0.011 Gbits/sec
[137] 0.0000-5.0170 sec 0.010 GBytes 0.017 Gbits/sec
[148] 0.0000-5.1156 sec 0.007 GBytes 0.011 Gbits/sec
[215] 0.0000-5.0418 sec 0.006 GBytes 0.010 Gbits/sec
[194] 0.0000-5.0388 sec 0.006 GBytes 0.011 Gbits/sec
[141] 0.0000-5.0138 sec 0.007 GBytes 0.012 Gbits/sec
[ 16] 0.0000-5.1256 sec 0.009 GBytes 0.015 Gbits/sec
[ 76] 0.0000-5.1248 sec 0.007 GBytes 0.011 Gbits/sec
[189] 0.0000-5.0621 sec 0.009 GBytes 0.015 Gbits/sec
[ 99] 0.0000-5.1593 sec 0.007 GBytes 0.012 Gbits/sec
[ 33] 0.0000-5.1374 sec 0.008 GBytes 0.013 Gbits/sec
[ 49] 0.0000-5.0833 sec 0.008 GBytes 0.013 Gbits/sec
[ 13] 0.0000-5.1415 sec 0.010 GBytes 0.017 Gbits/sec
[174] 0.0000-5.0306 sec 0.007 GBytes 0.012 Gbits/sec
[ 87] 0.0000-5.0579 sec 0.008 GBytes 0.014 Gbits/sec
[112] 0.0000-5.0512 sec 0.008 GBytes 0.014 Gbits/sec
[SUM] 0.0000-5.1957 sec 1.53 GBytes 2.53 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 0.194/0.725/17.330/1.980 ms (tot/err) = 215/0

```

Figure 20. Maximum satisfactory simulated connections.

Figure Figure 21 shows the test with 216 simulated connections, where it can be verified that the server cannot handle any more connections.

```

[ 89] 0.0000-5.2612 sec 0.011 GBytes 0.018 Gbits/sec
[143] 0.0000-5.2166 sec 0.008 GBytes 0.013 Gbits/sec
[ 33] 0.0000-5.0836 sec 0.010 GBytes 0.016 Gbits/sec
[115] 0.0000-5.2585 sec 0.009 GBytes 0.014 Gbits/sec
[ 74] 0.0000-5.0007 sec 0.005 GBytes 0.008 Gbits/sec
[197] 0.0000-5.3202 sec 0.010 GBytes 0.016 Gbits/sec
[ 34] 0.0000-5.2579 sec 0.006 GBytes 0.010 Gbits/sec
[ 71] 0.0000-5.3500 sec 0.008 GBytes 0.013 Gbits/sec
[118] 0.0000-5.2811 sec 0.003 GBytes 0.006 Gbits/sec
[  3] 0.0000-5.2609 sec 0.004 GBytes 0.007 Gbits/sec
[ 57] 0.0000-5.3103 sec 0.010 GBytes 0.017 Gbits/sec
[222] 0.0000-5.0367 sec 0.008 GBytes 0.013 Gbits/sec
[141] 0.0000-5.3680 sec 0.007 GBytes 0.011 Gbits/sec
[128] 0.0000-5.3463 sec 0.010 GBytes 0.015 Gbits/sec
[111] 0.0000-5.4001 sec 0.008 GBytes 0.012 Gbits/sec
[ 73] 0.0000-5.3746 sec 0.009 GBytes 0.014 Gbits/sec
[190] 0.0000-5.3597 sec 0.005 GBytes 0.007 Gbits/sec
[206] 0.0000-5.4225 sec 0.010 GBytes 0.016 Gbits/sec
[124] 0.0000-5.1335 sec 0.005 GBytes 0.009 Gbits/sec
[ 41] 0.0000-5.4476 sec 0.009 GBytes 0.014 Gbits/sec
[154] 0.0000-5.1716 sec 0.005 GBytes 0.008 Gbits/sec
[ 26] 0.0000-5.0086 sec 0.010 GBytes 0.018 Gbits/sec
[102] 0.0000-5.0192 sec 0.005 GBytes 0.008 Gbits/sec
[106] 0.0000-5.6502 sec 0.004 GBytes 0.006 Gbits/sec
[ 66] 0.0000-5.5587 sec 0.003 GBytes 0.005 Gbits/sec
[175] 0.0000-6.4137 sec 0.005 GBytes 0.006 Gbits/sec
[230] 0.0000-5.7672 sec 0.009 GBytes 0.013 Gbits/sec
[ 35] 0.0000-5.8569 sec 0.012 GBytes 0.018 Gbits/sec
[123] 0.0000-5.9575 sec 0.013 GBytes 0.018 Gbits/sec
[171] 0.0000-6.4363 sec 0.007 GBytes 0.010 Gbits/sec
[ 18] 0.0000-6.4749 sec 0.012 GBytes 0.016 Gbits/sec
[193] 0.0000-7.4261 sec 0.008 GBytes 0.009 Gbits/sec
[210] 0.0000-7.5097 sec 0.008 GBytes 0.009 Gbits/sec
client await server close failed: Resource temporarily unavailable
[201] 0.0000-5.1103 sec 0.000 GBytes 0.000 Gbits/sec
[SUM] 0.0000-5.2127 sec 1.47 GBytes 2.42 Gbits/sec
[ CT] final connect times (min/avg/max/stddev) = 0.192/0.438/19.294/1.383 ms (tot/err) = 216/0

```

Figure 21. Connection rejected.

In Figure 22, the resources were monitored while the tests were carried out with iperf, simulating the connection of users to the hotspot.

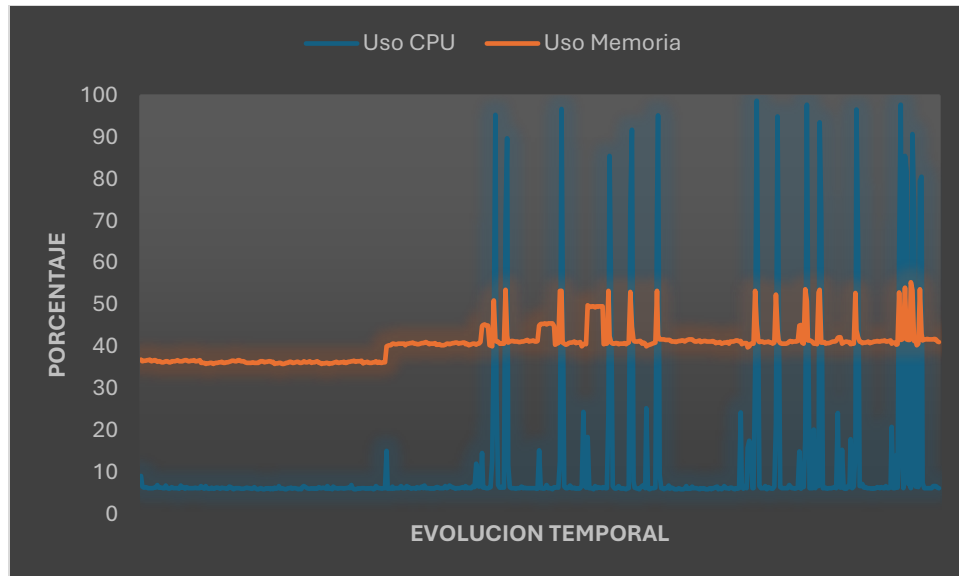


Figure 22. Resource monitoring with simulated users.

Elevation of Permits

Figure 23 shows password protection for permission elevation.

```
epn@raspberrypi:~ $ fupd
[sudo] password for epn:
Hit:1 http://archive.raspberrypi.org/debian bullseye InRelease
Hit:2 http://raspbian.raspberrypi.org/raspbian bullseye InRelease
Hit:3 https://ngrok-agent.s3.amazonaws.com buster InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  chromium-browser chromium-browser-l10n chromium-codecs-ffmpeg-extra
3 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Figure 23. Password-protected elevation of permissions.

Resource Testing with Glances

Figure 24 through Figure 30 present the monitoring of system resources using the Glances tool throughout the presentation, covering CPU consumption, RAM, network traffic, and disk usage.

Figure 24 shows the graph of resource and RAM consumption. Memory usage remains relatively constant at around 50%, with some minor fluctuations. On the other hand, CPU consumption reaches several high peaks as time goes on.

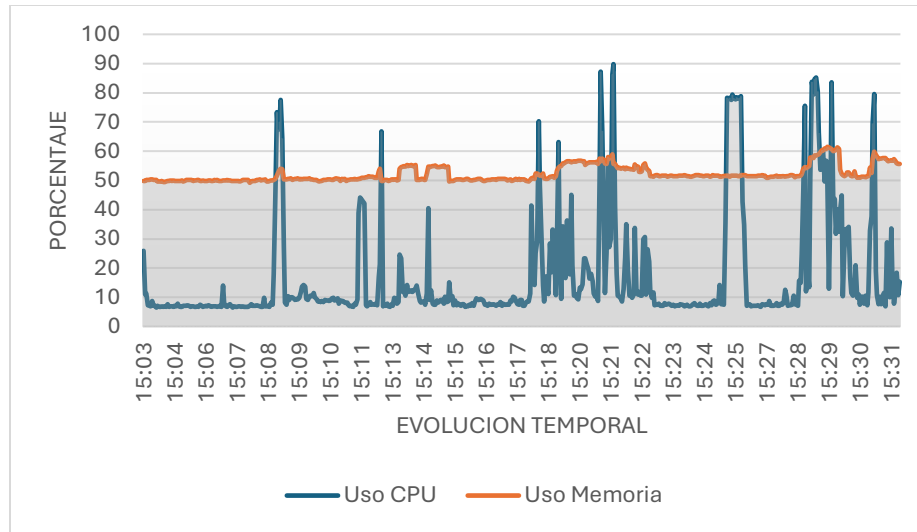


Figure 24. CPU and RAM consumption.

Figure 25 presents the maximum value of CPU and RAM usage during presentation, with CPU usage reaching a maximum level of 89.8% and memory usage reaching a maximum value of 61.5%.

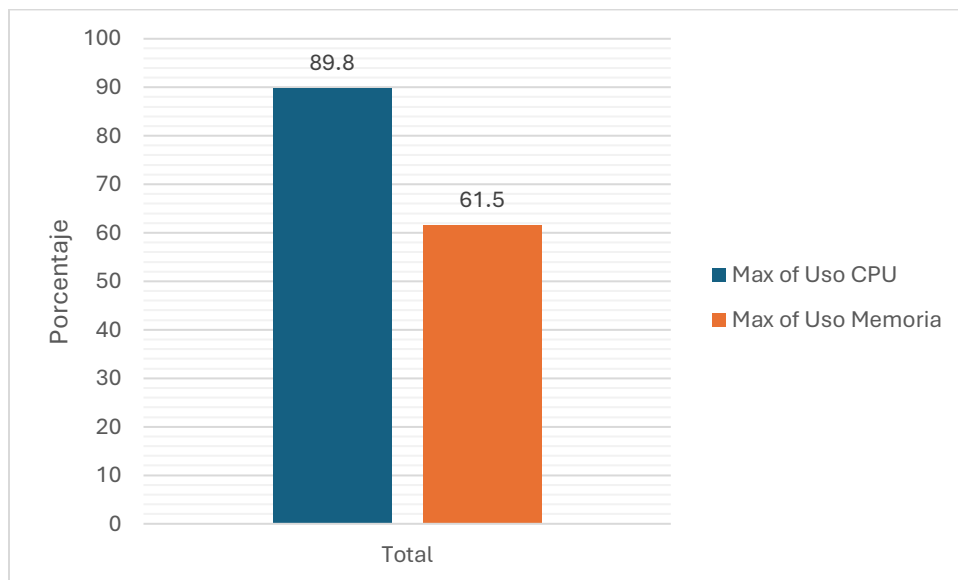


Figure 25. Maximum CPU and RAM consumption value.

Figure 26 shows the average usage of the 4 cores of the *Promedeux* device, verifying that none of them exceeds 25% usage.

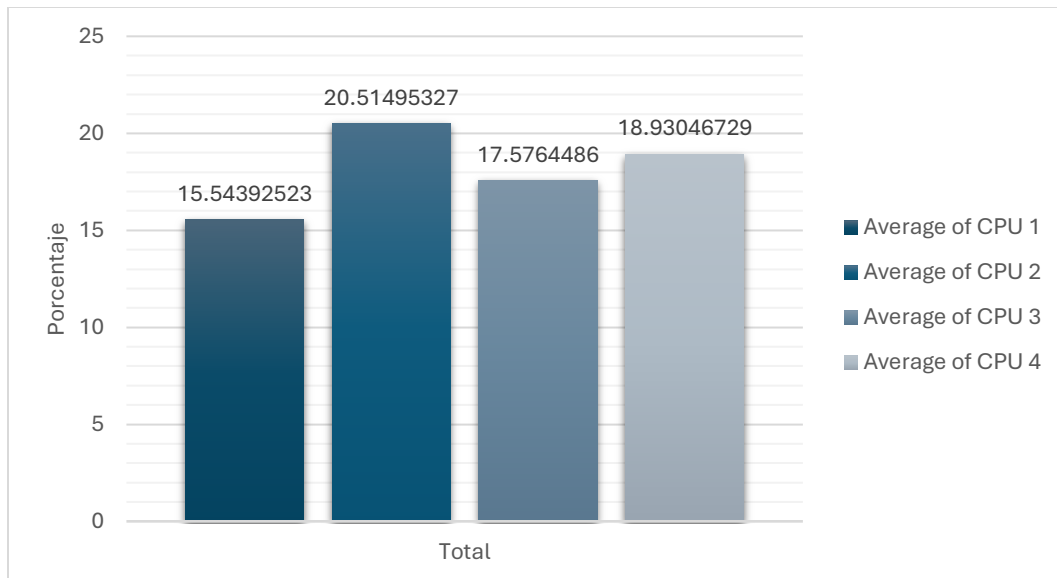


Figure 26. Average usage of the 4 CPU's of the Promedeux device.

Figure 27 represents the cumulative consumption of network traffic on the wlan0 and eth0 interfaces, checking a level of proportionality between the two interfaces.

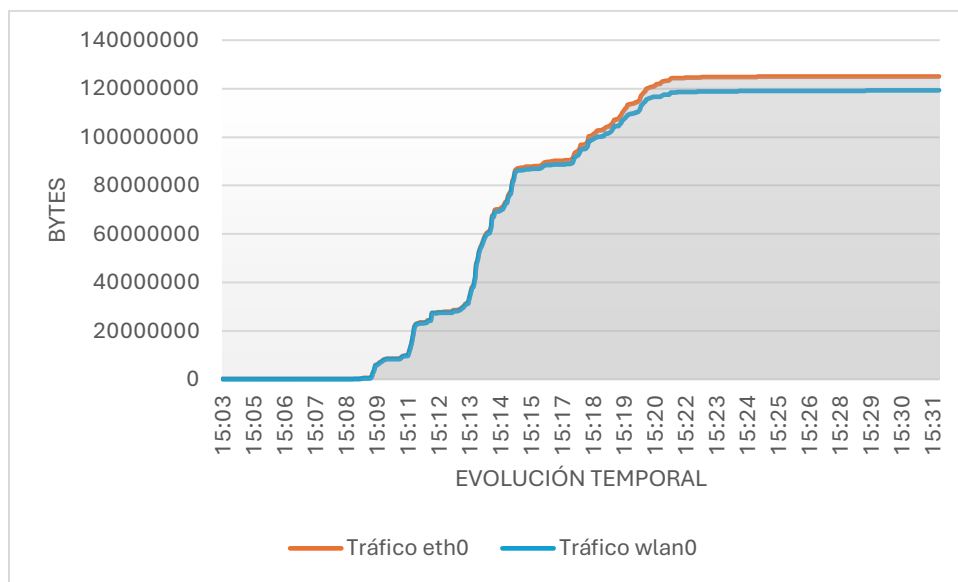


Figure 27. Cumulative consumption of network traffic.

Figure 28 presents the network consumption of the eth0 and wlan0 interfaces, showing a Gaussian bell-shaped trend.

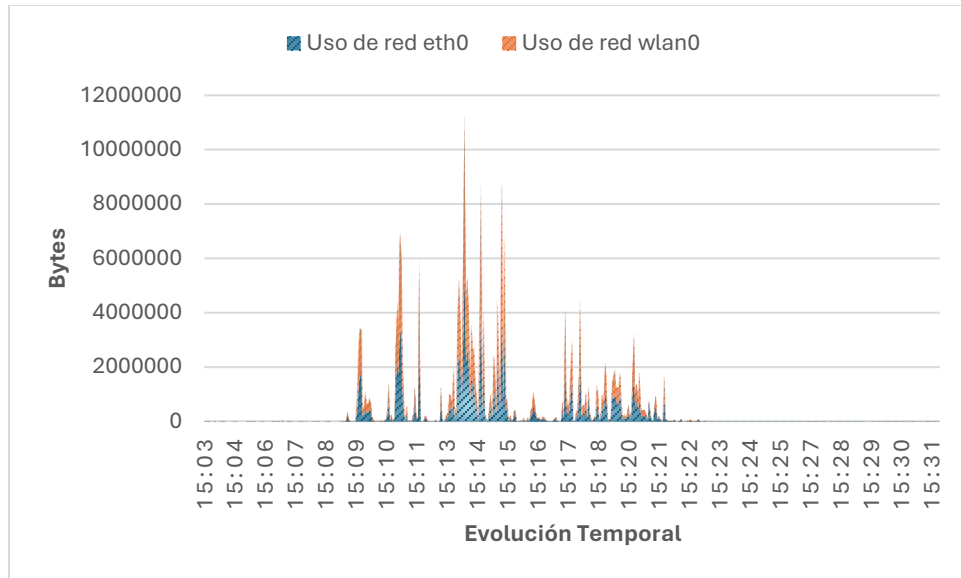


Figure 28. Network consumption of eth0 and wlan0 interfaces.

Figure 29 shows the use of disk in reading and writing throughout the presentation, evidencing areas where this resource is used intensively.

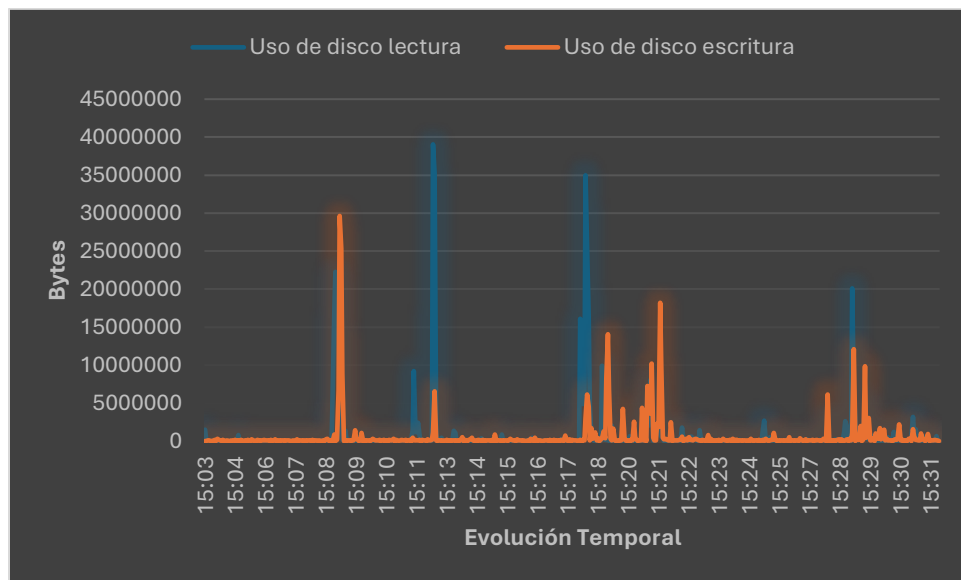


Figure 29. Disk usage in read and write.

Figure 30 illustrates the use of swap memory, a fraction of virtual memory created to improve interaction with the system.

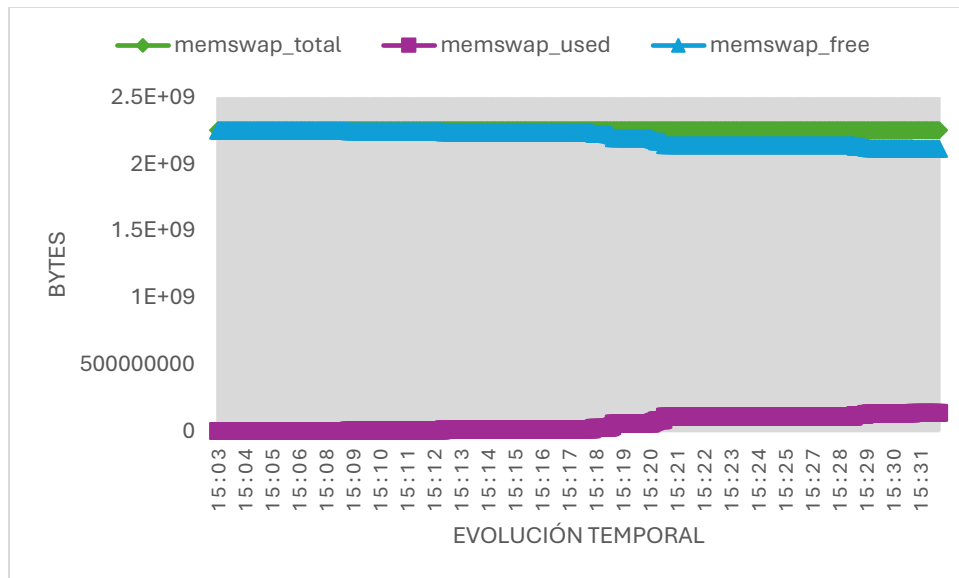


Figure 30. Swap memory usage.

Interpretation of the Results

Hotspot Connection

The results obtained in the hotspot connection scenario confirm the ability of the [Promedeux](#) device to act as an efficient wireless access point. The real-time association of devices with their respective IP and MAC addresses, and the ability to handle multiple users simultaneously, demonstrates that the device can support an educational environment with a considerable number of users. Although not all attendees interacted with the device, the results show that, with the correct configuration, an average of 15 customers can be handled. According to specialized forums, it has been proven that the hotspot can cover up to 32 simultaneous clients, providing a solid base for applications in classrooms, where commonly no more than 25 students are allowed.

Traffic Capture

Network traffic monitoring performed in the traffic capture scenario offers critical insight into the risks associated with unknown networks. The ability to identify visited domains and device IP addresses allows for detailed analysis of network behavior, essential for cybersecurity education. This demonstration significantly impacts users, highlighting the importance of browsing safety and the risk of sensitive data exposure.

Ngrok Service and Cloned Page

Ngrok's implementation to redirect traffic to a cloned page shows users' vulnerability to phishing attacks. The accessibility of the cloned page from a public link, and the possibility of accessing the resource even without an internet connection, demonstrate the versatility of the [Promedieux](#) device. This scenario is crucial for educating users about the importance of verifying the authenticity of the links and websites they visit, and how easily they can be tricked by fake pages.

POST Events

Capturing HTTP POST events on the cloned page evidences the system's ability to intercept and record critical data, such as user credentials. This hands-on exercise allows viewers to understand how cybercriminals obtain sensitive information using phishing techniques, underscoring the importance of being cautious when entering information into websites.

Firewall Rules and Network Stop

Manipulating firewall rules and stopping the network demonstrate the device's ability to control and manage network traffic. This functionality is vital for showing users how a denial-of-service attack can be carried out.

Brute Force Attack

The process of compressing and decompressing files using brute force attacks reveals the effectiveness and time required to crack passwords. This result is especially relevant in cybersecurity education, because it tangibly illustrates the importance of using strong and complex passwords, and the risks associated with weak passwords.

Dictionary Attack

Decompression of files using dictionary attacks highlights another effective password cracking technique. This method provides a practical comparison between different types of brute force and dictionary attacks, underscoring the need to educate users on creating strong passwords.

Keylogger

The implementation of a keylogger and keystroke capture illustrate how cybercriminals can monitor and record user activities. This scenario is critical to raising awareness of the dangers of the keylogger and the importance of security measures to protect personal information.

Iperf Testing and Elevation of Permits

Network bandwidth and stability tests using iperf demonstrate that the [Promedex](#) appliance can handle a significant user load without compromising performance. Password protection for permission elevation ensures that only authorized users can perform critical actions, providing an additional layer of security.

Resource Monitoring with Glances

Monitoring system resources using the Glances tool throughout the presentation provides a comprehensive view of device performance under different loads. Spikes in CPU and RAM consumption reflect the intensity of the tests, while the stability of average processor core usage suggests that the device can efficiently handle multiple tasks simultaneously. Analysis of network traffic and disk usage reveals the proportion and times of peak activity, crucial for assessing system capacity in high-demand scenarios.

Things to Detail

- Figure 26 shows that core usage reaches a level that provides a window large enough to prevent unexpected system shutdowns.
- Figure 28 shows a Gaussian bell-shaped trend. At the start of the presentation, there are no users connected and no hotspot activated. Traffic starts from the connection and

navigation of users and decreases due to the stop of the network and termination of scenarios that require connectivity and internet access.

- Figure 29 shows the trend of disk usage, which indicates that the highest usage occurs in the third and fourth scenarios, corresponding to brute force and dictionary attacks, as well as keylogger. This is confirmed when it is observed that, in the presentation, the use of disk in writing is more marked in the final parts of the graph.
- Figure 30 illustrates the use of swap memory, a fraction of virtual memory created to improve interaction with the system. You don't need to exhaust physical memory to activate swap memory; the system self-determines and uses it as needed. The graph shows the entire 2.1 GB memory and the inverse proportionality between the consumed and available memory. This resource becomes crucial by relieving the load on the hardware, avoiding overloading the processor.

Important Questions

Why is Table 1 showing a different password cracking time than Figure Figure 11, if it is the same word?

Table 1 shows several words that were compressed with the 4-character parameters and all character types (lowercase, uppercase, numbers, and all of the above). Figure Figure 9 shows the compression process in which 4 characters are used, but only lowercase characters are chosen as parameters. This is the reason for the variation: the algorithm saves the entered parameters and uses them for later decompression. When checking with all character types, more time is needed for decryption. The test considered this particularity, so it was carried out with the maximum number of characters and the most complex type to verify how long this action would take, being this the one that would require the most time. This facilitates the planning of the presentation, having adequate time for its execution.

Why does Figure 27 acquire a linear trend at the end?

It is observed that in the estimated period of time from 3:20 p.m. until the end of the presentation, the trend becomes stable. This is because the network has been stopped by using the `netstop` command, marking the end of the second scenario and the start of the third scenario.

These results highlight the ability of the **Promedeux** system to efficiently manage and analyze various aspects of cybersecurity in educational environments, providing a solid foundation for future implementations and improvements in cybersecurity education in Ecuador.